

Solving for angles in right triangles



1

a 82°

b 12°

c 56°

d 30°

e 45°

f 50°

g 61°

2

a $x = 38$

b $x = 68$

c $y = 47$

d $x = 70$

3 greater than 45°

4 $A = 66.4$

5 $t = 47$

6 $x = 53^\circ$

7

a $x = 29.74^\circ$

b $y = 150.26^\circ$

8 $A = 49$

9

a 30°

b $\frac{\sqrt{3}}{2}$

c $\frac{1}{\sqrt{3}}$

10

a 30°

b $\frac{1}{2}$

c $\frac{\sqrt{3}}{2}$

11

a i 30° ii $\frac{1}{2}$

b i 45° ii $\frac{1}{\sqrt{2}}$

c i 30° ii $\frac{1}{2}$

12

a i 60° ii $\sqrt{3}$

b i 45° ii 1

c i 60° ii $\sqrt{3}$

13 $\frac{24}{7}$



14

a Yes

b No

c No

d No

15 $\theta = 9.88$

16 $\theta = 54.78^\circ$

17 $\theta = 32.40$

18 $\theta = 49.12$

19 $y = 58.02$

20 $y = 54.90$

21

a $x = 33.31$

b $y = 6.97$

c $z = 14.39$

d 86 cm

22 $x = 22^\circ$

Additional questions

23

a 45°

b 64°

c 27°

d 16°

e 50°

f 83°

g 55°

h 45°

24

a 0°

b 45°

c 60°

d 65°

e 66°

f 66°

g 54°

h 53°

25

a 59.5°

b 61.2°

c 83.6°

d 7.1°

26

a $x = 74$

b $x = 69$



c $x = 74^\circ$

d $x = 74^\circ$

e $x = 38^\circ$

f $x = 53$

g $x = 62^\circ$

h $x = 67^\circ$

i $x = 67$

27 $x = 21.9^\circ, y = 48.8^\circ$

28

a $x = 26.57^\circ$

b $y = 50.19^\circ$

c $z = 13.24^\circ$

29

- a $\angle 1$ has the greater cosine value. Cosine is a ratio of the adjacent leg to the hypotenuse, and as the angle measure decreases, this ratio gets closer to 1 as the two sides of the triangle almost become the same length.
- b $\angle 2$ has the greater sine value. Sine is a ratio of the opposite leg to the hypotenuse, and both triangles have the same opposite leg, so the triangle with the shorter hypotenuse will have the greater value.
- c $\angle 2$ has the greater tangent value. Tangent is a ratio of the opposite leg to the adjacent leg, and both triangles have the same opposite leg, so the triangle with the shorter adjacent leg will have the greater value.